

IPB Lost & Found System

API Contract v1.0

Complete REST API Reference for Frontend Developers and AI Agents

Base URL (Production)	https://project-ads-lostn-found-system.vercel.app (frontend)
Base URL (Backend)	https://[railway-or-render-url]/ (FastAPI)
Auth Scheme	Bearer Token (JWT) — Header: Authorization: Bearer
Content-Type	application/json (multipart/form-data for file uploads)
API Version	1.0.0
Framework	FastAPI (Python) + PostgreSQL
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Table of Contents

1. Overview & Architecture

- 1.1 System Description
- 1.2 Role-Based Access Control
- 1.3 Authentication Flow
- 1.4 Common HTTP Status Codes
- 1.5 Pagination Parameters

2. Enumeration Reference

- 2.1 ItemType
- 2.2 ItemStatus
- 2.3 ItemCategory
- 2.4 ItemLocation
- 2.5 IPBLocation
- 2.6 ClaimStatus
- 2.7 FoundReportStatus
- 2.8 MatchStatus
- 2.9 UserRole
- 2.10 TimePeriod

3. Auth Endpoints /auth

- 3.1 POST /auth/login
- 3.2 GET /auth/me

4. Items Endpoints /items

- 4.1 GET /items — Public Item List
- 4.2 GET /items/admin/all — Admin Full List
- 4.3 GET /items/{item_id} — Item Detail
- 4.4 POST /items — Report Item
- 4.5 POST /items/{item_id}/foto — Upload Photo
- 4.6 POST /items/{item_id}/bukti-kepemilikan — Upload Ownership Proof
- 4.7 DELETE /items/{item_id} — Delete Item

5. Claims Endpoints /claims

- 5.1 POST /claims — Submit Claim
- 5.2 POST /claims/{claim_id}/bukti — Upload Evidence
- 5.3 GET /claims/me — My Claims
- 5.4 GET /claims — All Claims (Admin)
- 5.5 PATCH /claims/{claim_id}/approve — Approve Claim
- 5.6 PATCH /claims/{claim_id}/reject — Reject Claim
- 5.7 GET /claims/{claim_id}/kode — Get Pickup Code

6. Found Reports Endpoints /found-reports

- 6.1 POST /found-reports — Submit Found Report
- 6.2 POST /found-reports/{report_id}/foto — Upload Photo
- 6.3 GET /found-reports/me — My Reports
- 6.4 GET /found-reports — All Pending Reports (Admin)
- 6.5 GET /found-reports/{report_id}/kode — Get Pickup Code
- 6.6 PATCH /found-reports/{report_id}/approve — Approve Report
- 6.7 PATCH /found-reports/{report_id}/reject — Reject Report

7. Matches Endpoints /matches

- 7.1 GET /matches — List Pending Matches
- 7.2 PATCH /matches/{match_id}/confirm — Confirm Match
- 7.3 PATCH /matches/{match_id}/reject — Reject Match

8. Notifications Endpoints /notifications

- 8.1 GET /notifications — Get My Notifications
- 8.2 PATCH /notifications/read — Mark All as Read

9. Leaderboard Endpoints /leaderboard

- 9.1 GET /leaderboard — Get Leaderboard
- 9.2 GET /leaderboard/me/riwayat — My Points History

10. Data Models Reference

- 10.1 User
- 10.2 Item
- 10.3 Claim
- 10.4 FoundReport
- 10.5 ItemMatch
- 10.6 Notification
- 10.7 PointLog

11. Business Logic & Rules

- 11.1 Points System
- 11.2 Auto-Match Engine
- 11.3 Item Expiry
- 11.4 Pickup Codes

1. Overview & Architecture

1.1 System Description

IPB Lost & Found is a web platform for IPB University students and administrators to manage lost and found items across campus. Students (mahasiswa) can report lost/found items, submit ownership claims, and earn points. Admins review claims, approve/reject matches, and manage the item lifecycle. The backend is a FastAPI REST API with PostgreSQL database.

1.2 Role-Based Access Control (RBAC)

Role	Description	Access Level
Guest	Unauthenticated visitor	Can browse public item list and leaderboard only
mahasiswa	Authenticated student with NIM	Report items, submit claims, upload evidence, view own data, earn points
admin	System administrator	Full access: approve/reject claims & reports, manage matches, view all data, delete items

1.3 Authentication Flow

1. Call **POST /auth/login** with username & password.
2. Receive **access_token** (JWT) in the response.
3. Include token in every protected request header:

Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

Token contains: **sub** (username), **exp** (expiry). Algorithm: HS256.

1.4 Common HTTP Status Codes

Code	Meaning	Typical Scenario
200 OK	Success	GET request returned data successfully
201 Created	Resource created	POST created a new item/claim/report
400 Bad Request	Validation error	Missing required field or invalid enum value
401 Unauthorized	Not authenticated	No/invalid/expired Bearer token
403 Forbidden	Insufficient role	Mahasiswa accessing admin-only endpoint
404 Not Found	Resource not found	item_id, claim_id, report_id not found in DB
422 Unprocessable	Schema error	Invalid field type or enum value in request body
500 Internal Error	Server error	Unexpected error (report to dev team)

1.5 Pagination Query Parameters

All list endpoints support the following query parameters:

Field	Type	Required	Description
skip	integer	Optional	Number of records to skip. Default: 0. Minimum: 0
limit	integer	Optional	Max records to return. Default: 24. Range: 1–100

Example: GET /items?skip=24&limit=12 — retrieves page 2 with 12 items per page.

2. Enumeration Reference

All enum fields must use the exact string values listed below. Sending an invalid enum value will return HTTP 422 Unprocessable Entity.

2.1 ItemType

Defines whether the item is a lost item report or a found item report.

Enum: ItemType	Valid Values
ItemType	"LOST", "FOUND"

2.2 ItemStatus

OPEN: active and searchable. PENDING: has an active claim/report under review. CLOSED: resolved. EXPIRED: no activity for 30 days.

Enum: ItemStatus	Valid Values
ItemStatus	"OPEN", "PENDING", "CLOSED", "EXPIRED"

2.3 ItemCategory

Category of the item. Used for filtering and auto-match scoring.

Enum: ItemCategory	Valid Values
ItemCategory	"elektronik", "dompet", "kunci", "kartu", "pakaian", "tas", "botol", "lainnya"

2.4 TimePeriod

Time filter for item listing. Filters by tanggal (incident date).

Enum: TimePeriod	Valid Values
TimePeriod	"today", "this_week", "this_month", "all_time"

2.5 ClaimStatus

Status of an ownership claim submitted by a mahasiswa.

Enum: ClaimStatus	Valid Values
ClaimStatus	"PENDING", "APPROVED", "REJECTED"

2.6 FoundReportStatus

Status of a found-item report submitted by a mahasiswa.

Enum: FoundReportStatus	Valid Values
FoundReportStatus	"PENDING", "APPROVED", "REJECTED"

2.7 MatchStatus

Status of an auto-detected match between a FOUND item and a LOST item.

Enum: MatchStatus	Valid Values
MatchStatus	"PENDING", "CONFIRMED", "REJECTED"

2.8 UserRole

Role determines which endpoints are accessible.

Enum: UserRole	Valid Values
UserRole	"mahasiswa", "admin"

2.9 ItemLocation (lokasi_ditemukan_list / lokasi_kemungkinan_list)

Campus area locations — used for structured location tagging to enable auto-match.

"Common Class Room (CCR)" | "Gedung Kuliah Bersama (GKB)" | "Perpustakaan Pusat" | "Masjid Al-Hurriyyah" | "Gymnasium" | "Auditorium Andi Hakim Nasoetion" | "Asrama Putra (Astra)" | "Asrama Putri (Astri)" | "Kantin Stevia" | "Kantin Ungu" | "Blue Corner" | "Yellow Corner" | "Kantin Nays" | "Kantin Rimbawan" | "Kantin Sapta" | "Kantin Plasma" | "Kantin Empat (Kanpat)" | "Kantin Ibu Sayang" | "Kantin Makjan" | "Fakultas Pertanian (Faperta)" | "Fakultas Kedokteran Hewan (FKH)" | "Fakultas Perikanan dan Ilmu Kelautan (FPIK)" | "Fakultas Peternakan (Fapet)" | "Fakultas Kehutanan dan Lingkungan (Fahutan)" | "Fakultas Teknologi Pertanian (Fateta)" | "Fakultas Matematika dan Ilmu Pengetahuan Alam (FMIPA)" | "Fakultas Ekonomi dan Manajemen (FEM)" | "Fakultas Ekologi Manusia (FEMA)" | "Sekolah Kedokteran Hewan dan Biomedis (SKHB)" | "Sekolah Bisnis (SB)" | "Sekolah Vokasi (SV)" | "Halte Bus / Lintas" | "Area Jalanan Kampus" | "Lainnya"

2.10 IPBLocation (lokasi_sekarang)

Specific handover/storage locations on campus. Used for FOUND items to indicate where the item is currently being kept (e.g., security post, admin desk).

"Pos Satpam Faperta" | "Pos Satpam SKHB" | "Pos Satpam FPIK" | "Pos Satpam Fapet" | "Pos Satpam Fahutan" | "Pos Satpam Fateta" | "Pos Satpam FMIPA" | "Pos Satpam FEM" | "Pos Satpam Fema" | "Meja Informasi CCR Lantai 1" | "Meja Sirkulasi Perpustakaan Pusat" | "Lobi Utama Gedung Kuliah Bersama (GKB)" | "Sekretariat Masjid Al-Hurriyyah" | "Pos Keamanan Pintu Utama GWW" | "Ruang Pengelola Gymnasium" | "Resepsionis Klinik IPB Dramaga" | "Pos Pengamanan Lobi Rektorat AHN" | "Sekretariat BEM KM IPB (Student Center)" | "Kantor Pengelola Asrama PKU Putra" | "Kantor Pengelola Asrama PKU Putri" | "Pos Penjagaan Asrama Sylvapinus" | "Shelter Bus Kampus Rektorat"

3. Auth Endpoints /auth

3.1 Login

POST**/auth/login**

■ Public (no auth required)

Authenticates a user and returns a JWT access token.

Request Body (JSON):

Field	Type	Required	Description
username	string	✓ Required	The user's username
password	string	✓ Required	The user's password (plaintext, hashed server-side)

Request Example:

```
{
  "username": "budi_santoso",
  "password": "mypassword123"
}
```

Response 200 OK:

Field	Type	Description
access_token	string	JWT token to use in Authorization header
token_type	string	Always "bearer"
role	UserRole	"mahasiswa" or "admin"
nama	string	User's display name

Response Example:

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "role": "mahasiswa",
  "nama": "Budi Santoso"
}
```

Error Responses:

```
401 — Credentials invalid (wrong username or password)
```

3.2 Get Current User Profile

GET**/auth/me**

■ Any logged-in user

Returns the profile of the currently authenticated user.

Response 200 OK:

Field	Type	Description
id	integer	User's unique ID
username	string	User's username
nim	string null	Student NIM (null for admin)
nama	string	User's full name
email	string	User's email address
role	UserRole	"mahasiswa" or "admin"
poin	integer	Total accumulated points

4. Items Endpoints /items

Items are the core entity of the system. An item can be either LOST (barang hilang reported by owner) or FOUND (barang temuan reported by finder). Each has different required fields. **Privacy rule:** Public responses (ItemOutPublik) hide: deskripsi_detail, lokasi_sekarang, bukti_kepemilikan. Full responses (ItemOut) are returned only to the owner, the finder, or admin.

4.1 List Items (Public)

GET

/items

■ Public (optional auth)

Returns a paginated list of items. Supports filtering by type, category, location, status, and search. If authenticated, the response includes photos for FOUND items. Unauthenticated guests see items without photos.

Query Parameters:

Field	Type	Required	Description
tipe	ItemType	Optional	Filter by type: LOST or FOUND
kategori	ItemCategory	Optional	Filter by category
lokasi	ItemLocation	Optional	Filter by area location (lokasi_ditemukan_list or lokasi_kemungkinan_list)
status	ItemStatus	Optional	Filter by status: OPEN, PENDING, CLOSED, EXPIRED
q	string	Optional	Free-text search on nama_publik
period	TimePeriod	Optional	Time filter: today, this_week, this_month, all_time
skip	integer	Optional	Pagination offset. Default: 0
limit	integer	Optional	Page size. Default: 24, max: 100

Response 200 OK — Array of ItemOutPublik:

Field	Type	Description
id	integer	Item unique ID
user_id	integer	ID of reporter (used by frontend to check isOwner)
tipe	ItemType	LOST or FOUND
status	ItemStatus	Current status
kategori	ItemCategory	Item category
nama_publik	string	Generic public name (e.g. 'Dompét Hitam')
lokasi_ditemukan	string null	FOUND only: free-text where item was found
lokasi_kemungkinan	string[] null	LOST only: array of probable locations (free-text)
lokasi_ditemukan_list	ItemLocation null	FOUND only: structured location enum
lokasi_kemungkinan_list	ItemLocation[] null	LOST only: structured location enum array
tanggal	datetime	Incident date (ISO 8601)
expired_at	datetime null	When the item expires (30 days after creation)

Field	Type	Description
foto	ItemFotoOut[]	Array of photo objects: {id: int, url: string}. Empty for guests on FOUND items.

4.2 Admin Full Item List

GET /items/admin/all

■ Admin Only

Returns all items with full details including sensitive fields. Supports same query params as 4.1.

Response 200 OK — Array of ItemOut (full detail):

Field	Type	Description
id	integer	Item ID
user_id	integer	Reporter's user ID
tipe	ItemType	LOST or FOUND
status	ItemStatus	Current status
kategori	ItemCategory	Category
nama_public	string	Generic public name
deskripsi_detail	string null	Sensitive detail — only visible to admin/owner
lokasi_ditemukan	string null	FOUND: where found (free-text)
lokasi_sekarang	IPBLocation null	FOUND: current storage location (sensitive)
lokasi_kemungkinan	string[] null	LOST: probable locations (free-text)
bukti_kepemilikan	string[] null	LOST: URLs of ownership proof photos (sensitive)
lokasi_ditemukan_list	ItemLocation null	Structured location for FOUND
lokasi_kemungkinan_list	ItemLocation[] null	Structured locations for LOST
tanggal	datetime	Incident date
expired_at	datetime null	Expiry datetime
created_at	datetime	Record creation time
foto	ItemFotoOut[]	Photos: [{id, url}]

4.3 Item Detail

GET /items/{item_id}

■ Public (optional auth)

Returns item detail. Response schema depends on caller:

• **Admin or Owner:** Returns full ItemOut (includes deskripsi_detail, lokasi_sekarang, bukti_kepemilikan) • **Other authenticated users:** Returns ItemOutPublik (sensitive fields hidden) • **Guest:** Returns ItemOutPublik (photos hidden for FOUND items)

Path Parameter:

Field	Type	Required	Description
item_id	integer	✓ Required	Unique ID of the item

4.4 Report / Create Item

POST	/items	■ Mahasiswa Only
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Creates a new lost or found item report. After creation, the system automatically runs the auto-match engine to find potential matches.

Request Body (JSON) — ItemCreate:

Field	Type	Required	Description
tipe	ItemType	✓ Required	LOST or FOUND
kategori	ItemCategory	✓ Required	Item category
nama_publik	string	✓ Required	Generic public name (do NOT include identifying details)
deskripsi_detail	string	Optional	Sensitive details only visible to admin/owner
tanggal	datetime	✓ Required	Date/time of incident (ISO 8601)
lokasi_ditemukan	string	Required if FOUND	Free-text where item was found
lokasi_sekarang	IPBLocation	Required if FOUND	Current storage location enum
lokasi_ditemukan_list	ItemLocation	Optional	Structured location for FOUND (for matching)
lokasi_kemungkinan	string[]	Required if LOST	Array of probable locations (free-text)
lokasi_kemungkinan_list	ItemLocation[]	Optional	Structured locations for LOST (for matching)

FOUND Item Example:

{
"tipe": "FOUND",
"kategori": "dompet",
"nama_publik": "Dompet Hitam",
"deskripsi_detail": "Dompet kulit hitam berisi KTM dan uang 50rb",
"lokasi_ditemukan": "Dekat parkir FEM lantai 1",
"lokasi_sekarang": "Pos Satpam FEM",
"lokasi_ditemukan_list": "Fakultas Ekonomi dan Manajemen (FEM)",
"tanggal": "2025-05-30T10:00:00"
}

LOST Item Example:

{
"tipe": "LOST",

```

    "kategori": "elektronik",
    "nama_publik": "Earphone TWS Putih",
    "deskripsi_detail": "TWS merek Samsung Galaxy Buds, case hitam, ada stiker bintang",
    "lokasi_kemungkinan": ["Kantin Stevia", "Perpustakaan Pusat"],
    "lokasi_kemungkinan_list": ["Kantin Stevia", "Perpustakaan Pusat"],
    "tanggal": "2025-05-29T14:30:00"
  }

```

Response 200 OK: Full ItemOut object (see section 4.2 for schema).

4.5 Upload Item Photo

POST	<code>/items/{item_id}/foto</code>	■ Mahasiswa Only (owner)
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Uploads a photo for an item. Only the item's owner can upload.

Content-Type: multipart/form-data

Field	Type	Required	Description
item_id	integer (path)	✓ Required	Item ID to attach photo to
file	File (form-data)	✓ Required	Image file (JPEG, PNG, etc.)

Response 200 OK: Full ItemOut with updated foto array.

4.6 Upload Ownership Proof

POST	<code>/items/{item_id}/bukti-kepemilikan</code>	■ Mahasiswa Only (owner)
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Uploads ownership proof photo for a LOST item (e.g., photo showing item with owner, receipt, etc.). Stored in bukti_kepemilikan field, visible only to admin/owner.

Content-Type: multipart/form-data

Field	Type	Required	Description
item_id	integer (path)	✓ Required	ID of the LOST item
file	File (form-data)	✓ Required	Image file (proof of ownership)

Response 200 OK: Full ItemOut with updated bukti_kepemilikan list.

4.7 Delete Item

DELETE	<code>/items/{item_id}</code>	■ Admin Only
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Permanently deletes an item and all associated data (claims, matches, photos).

Field	Type	Required	Description
item_id	integer (path)	✓ Required	ID of the item to delete

Response 200 OK:

```

{"message": "Barang berhasil dihapus"}

```

5. Claims Endpoints /claims

A Claim is submitted by a mahasiswa who believes a FOUND item belongs to them. The claim includes a description of distinguishing features (ciri khusus) and optionally photo evidence. Admin reviews and either approves or rejects. On approval, a pickup code is generated.

5.1 Submit Claim

POST

/claims

■ Mahasiswa Only

Submits a claim for a FOUND item. The mahasiswa must describe unique identifying features not visible in the public listing (ciri khusus).

Request Body (JSON):

Field	Type	Required	Description
item_id	integer	✓ Required	ID of the FOUND item being claimed
deskripsi_ciri	string	✓ Required	Description of unique features not visible in public photos
{			
"item_id": 42,			
"deskripsi_ciri": "Di dalam dompet ada kartu ATM BRI atas nama Budi, struk belanja, dan foto keluarga kecil"			
}			

Response 200 OK — ClaimOut:

Field	Type	Description
id	integer	Claim unique ID
item_id	integer	Item being claimed
user_id	integer	Claimant's user ID
status	ClaimStatus	PENDING APPROVED REJECTED
deskripsi_ciri	string	Submitted description of features
bukti_foto	string[]	URLs of evidence photos
created_at	datetime	Submission timestamp
catatan_admin	string null	Admin's approval/rejection note
kode_pengambilan	string null	Pickup code (only set when APPROVED)

5.2 Upload Claim Evidence Photo

POST

/claims/{claim_id}/bukti

■ Mahasiswa Only (claimant)

Adds a photo as evidence to support the ownership claim.

Content-Type: multipart/form-data

Field	Type	Required	Description
claim_id	integer (path)	✓ Required	ID of the claim
file	File (form-data)	✓ Required	Evidence photo file

Response 200 OK: Updated ClaimOut with new photo in bukti_foto array.

5.3 Get My Claims

GET /claims/me

■ Mahasiswa Only

Returns all claims submitted by the authenticated mahasiswa.

Query Params: skip, limit (see Section 1.5).

Response 200 OK: Array of ClaimOut.

5.4 Get All Claims (Admin)

GET /claims

■ Admin Only

Returns all claims in the system for admin review.

Query Params: skip, limit (see Section 1.5).

Response 200 OK: Array of ClaimOut.

5.5 Approve Claim

PAT CH /claims/{claim_id}/approve

■ Admin Only

Approves a claim. This action:

1. Sets claim status to APPROVED
2. Generates a unique 6-character pickup code (kode_pengambilan)
3. Sets the item status to CLOSED
4. Awards points to the finder (+10 poin)
5. Sends notifications to both claimant and item reporter

Query Parameters:

Field	Type	Required	Description
claim_id	integer (path)	✓ Required	ID of the claim to approve
catatan	string (query)	Optional	Admin note for the claimant. Default: empty string

Response 200 OK: Updated ClaimOut with status=APPROVED and kode_pengambilan set.

5.6 Reject Claim

PAT CH /claims/{claim_id}/reject

■ Admin Only

Rejects a claim. Item status returns to OPEN.

Field	Type	Required	Description
claim_id	integer (path)	✓ Required	ID of the claim to reject
catatan	string (query)	Optional	Reason for rejection shown to claimant. Default: empty string

Response 200 OK: Updated ClaimOut with status=REJECTED.

5.7 Get Claim Pickup Code

GET	/claims/{claim_id}/kode	■ Admin Only
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Returns the pickup code for an approved claim. Used during physical item handover.

Field	Type	Required	Description
claim_id	integer (path)	✓ Required	ID of the approved claim

Response 200 OK:

Field	Type	Description
claim_id	integer	Claim ID
nama_barang	string	Public name of the claimed item
kode_pengambilan	string null	6-character pickup code (null if not yet approved)
status	ClaimStatus	Current claim status

6. Found Reports Endpoints /found-reports

A Found Report is submitted by a mahasiswa who physically found a lost item that was previously reported. Unlike a Claim (owner proving ownership), a Found Report is a finder saying 'I found this lost item and I have it here'. Admin verifies and generates a pickup code for the original item owner.

6.1 Submit Found Report

POST

/found-reports

■ Mahasiswa Only

Reports that a LOST item has been found by the submitter. Can optionally include a photo. Uses multipart/form-data.

Content-Type: multipart/form-data

Field	Type	Required	Description
lost_item_id	integer (form)	✓ Required	ID of the LOST item that was found
deskripsi	string (form)	✓ Required	Description of how/where the item was found
lokasi_sekarang	IPBLocation (form)	✓ Required	Where the item is currently being kept
foto	File (form)	Optional	Optional photo evidence of the found item

Response 200 OK — FoundReportOut:

Field	Type	Description
id	integer	Report unique ID
lost_item_id	integer	ID of the referenced LOST item
lost_item_nama	string null	Public name of the lost item
reporter_id	integer	ID of the mahasiswa who found it
reporter_nama	string null	Name of the finder
deskripsi	string	Description submitted by finder
lokasi_sekarang	IPBLocation	Current storage location
foto_bukti	string[]	URLs of evidence photos
status	FoundReportStatus	PENDING APPROVED REJECTED
catatan_admin	string null	Admin note
kode_pengambilan	string null	Pickup code (set on approval)
created_at	datetime	Submission time

6.2 Upload Found Report Photo

POST

/found-reports/{report_id}/foto

■ Mahasiswa Only (reporter)

Adds a photo to an existing found report.

Content-Type: multipart/form-data

Field	Type	Required	Description
report_id	integer (path)	✓ Required	ID of the found report
file	File (form-data)	✓ Required	Evidence photo

Response 200 OK: Updated FoundReportOut.

6.3 Get My Found Reports

GET /found-reports/me

■ Mahasiswa Only

Returns all found reports submitted by the authenticated mahasiswa.

Query Params: skip, limit.

Response 200 OK: Array of FoundReportOut.

6.4 Get All Pending Found Reports (Admin)

GET /found-reports

■ Admin Only

Returns all found reports with status PENDING, for admin review.

Query Params: skip, limit.

Response 200 OK: Array of FoundReportOut.

6.5 Get Found Report Pickup Code

GET /found-reports/{report_id}/kode

■ Admin Only

Returns the pickup code for an approved found report.

Field	Type	Required	Description
report_id	integer (path)	✓ Required	ID of the found report

Field	Type	Description
report_id	integer	Report ID
nama_barang	string	Name of the found/lost item
kode_pengambilan	string null	6-character pickup code (null if not approved)
status	FoundReportStatus	Current report status

6.6 Approve Found Report

PATCH /found-reports/{report_id}/approve

■ Admin Only

Approval actions: 1. Sets report status to APPROVED 2. Generates a unique pickup code for the item owner 3. Sets the LOST item's status to CLOSED 4. Awards +10 points to the finder (reporter) 5. Sends notification to the item owner and the finder

Field	Type	Required	Description
report_id	integer (path)	✓ Required	ID of the found report to approve

Field	Type	Required	Description
catatan	string (query)	Optional	Admin note. Default: empty string

Response 200 OK: Updated FoundReportOut with status=APPROVED.

6.7 Reject Found Report

PATCH	/found-reports/{report_id}/reject	■ Admin Only
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Rejects a found report.

Field	Type	Required	Description
report_id	integer (path)	✓ Required	ID of the found report to reject
catatan	string (query)	Optional	Rejection reason. Default: empty string

Response 200 OK: Updated FoundReportOut with status=REJECTED.

7. Matches Endpoints /matches

The auto-match engine runs automatically every time a new item is reported. It compares the new item against existing items of the opposite type (FOUND vs LOST) and scores them based on: same category (+3pts) and overlapping campus location (+2pts). If score ≥ 3 , a match is created with status PENDING. Admin then reviews and confirms or rejects.

MatchOut Schema:

Field	Type	Description
id	integer	Match unique ID
found_item_id	integer	ID of the FOUND item
lost_item_id	integer	ID of the LOST item
alasan_match	string	System-generated explanation (e.g. 'Kategori sama: dompet Kecocokan Lokasi: FEM')
status	MatchStatus	PENDING CONFIRMED REJECTED
catatan_admin	string null	Admin note when confirming/rejecting
created_at	datetime	When the match was detected

7.1 List Pending Matches

GET /matches

■ Admin Only

Returns all matches with status PENDING, waiting for admin review.

Query Params: skip, limit.

Response 200 OK: Array of MatchOut.

7.2 Confirm Match

PATCH /matches/{match_id}/confirm

■ Admin Only

Admin confirms that the auto-detected match is correct.

Confirmation actions: 1. Sets match status to CONFIRMED 2. Sets both FOUND and LOST items to CLOSED 3. Awards +10 points to the FOUND item reporter (finder) 4. Sends notifications to both item owners (found+lost)

Field	Type	Required	Description
match_id	integer (path)	✓ Required	ID of the match to confirm
catatan	string (query)	Optional	Admin note. Default: empty string

Response 200 OK: Updated MatchOut with status=CONFIRMED.

7.3 Reject Match

PATCH /matches/{match_id}/reject

■ Admin Only

Admin rejects the auto-detected match (false positive). Items return to OPEN.

Field	Type	Required	Description
match_id	integer (path)	✓ Required	ID of the match to reject
catatan	string (query)	Optional	Rejection reason. Default: empty string

Response 200 OK: Updated MatchOut with status=REJECTED.

8. Notifications Endpoints /notifications

Notifications are automatically created by the system when important events occur: claim approved/rejected, found report approved/rejected, match confirmed/rejected. Each user sees only their own notifications.

NotifOut Schema:

Field	Type	Description
id	integer	Notification unique ID
judul	string	Notification title
pesan	string	Full notification message
is_read	boolean	Whether the user has read this notification
created_at	datetime	When notification was created

8.1 Get My Notifications

GET

/notifications

■ Any logged-in user

Returns all notifications for the authenticated user, ordered by newest first.

Query Params: skip, limit.

Response 200 OK: Array of NotifOut.

8.2 Mark All Notifications as Read

PAT
CH

/notifications/read

■ Any logged-in user

Marks all of the user's notifications as read (is_read = true).

Response 200 OK:

```
{"message": "Semua notifikasi ditandai sudah dibaca"}
```

9. Leaderboard Endpoints /leaderboard

The leaderboard ranks mahasiswa by their total points. Points are earned by reporting found items and having them verified. Anyone (including guests) can view the leaderboard. Point history is private to each mahasiswa.

9.1 Get Leaderboard

GET

/leaderboard

■ Public (optional auth)

Returns the top 10 mahasiswa by points. If authenticated, also returns caller's rank and points.

Response 200 OK — LeaderboardResponse:

Field	Type	Description
top10	LeaderboardEntry[]	Array of top 10 entries
top10[].peringkat	integer	Rank (1 = highest)
top10[].user_id	integer	User ID
top10[].nama	string	User's full name
top10[].poin	integer	Total points
poin_saya	integer null	Caller's total points (null if not authenticated)
peringkat_saya	integer null	Caller's rank (null if not authenticated or not ranked)

Response Example:

```
{
  "top10": [
    {"peringkat": 1, "user_id": 5, "nama": "Siti Rahayu", "poin": 120},
    {"peringkat": 2, "user_id": 12, "nama": "Ahmad Fauzi", "poin": 90}
  ],
  "poin_saya": 45,
  "peringkat_saya": 7
}
```

9.2 Get My Points History

GET

/leaderboard/me/riwayat

■ Mahasiswa Only

Returns the authenticated mahasiswa's complete point transaction history.

Response 200 OK — Array of PointLogOut:

Field	Type	Description
id	integer	Log entry ID
jumlah	integer	Points earned (positive = earned, negative = deducted)
alasan	string	Reason for the point transaction

Field	Type	Description
created_at	string (datetime)	When the points were awarded

10. Data Models Reference

Complete database entity definitions for reference.

10.1 User

Field	Type	Description
id	integer (PK)	Auto-incrementing primary key
username	string (unique)	Login identifier
nim	string (unique, nullable)	Student ID number (null for admins)
nama	string	Full display name
email	string (unique)	Email address
password	string	Bcrypt-hashed password (never returned in API)
role	UserRole	mahasiswa or admin
is_active	boolean	Account active status. Default: true
poin	integer	Accumulated points. Default: 0
created_at	datetime	Account creation timestamp

10.2 Item

Field	Type	Description
id	integer (PK)	Primary key
user_id	integer (FK → users)	Reporter's user ID
tipe	ItemType	LOST or FOUND
status	ItemStatus	OPEN PENDING CLOSED EXPIRED. Default: OPEN
kategori	ItemCategory	Item category
nama publik	string	Generic public name
deskripsi_detail	text (nullable)	Sensitive details (admin/owner only)
lokasi_ditemukan	string (nullable)	FOUND: free-text location where found
lokasi_sekarang	IPBLocation (nullable)	FOUND: current storage location
lokasi_ditemukan_list	ItemLocation (nullable)	FOUND: structured location enum
lokasi_kemungkinan	string[] (nullable)	LOST: array of probable locations (free-text)
lokasi_kemungkinan_list	ItemLocation[] (nullable)	LOST: structured location enum array

Field	Type	Description
bukti_kepemilikan	string[] (nullable)	LOST: ownership proof photo URLs
tanggal	datetime	Incident date/time
expired_at	datetime	Expiry time. Auto-set to created_at + 30 days
created_at	datetime	Record creation time

10.3 Claim

Field	Type	Description
id	integer (PK)	Primary key
item_id	integer (FK → items)	The FOUND item being claimed
user_id	integer (FK → users)	The mahasiswa submitting the claim
status	ClaimStatus	PENDING APPROVED REJECTED. Default: PENDING
deskripsi_ciri	text	Description of unique identifying features
bukti_foto	string[]	Array of evidence photo URLs. Default: []
catatan_admin	text (nullable)	Admin's approval/rejection note
kode_pengambilan	string (nullable)	6-char pickup code, set on approval
created_at	datetime	Submission time
updated_at	datetime (nullable)	Last modification time

10.4 FoundReport

Field	Type	Description
id	integer (PK)	Primary key
lost_item_id	integer (FK → items)	The LOST item that was found
reporter_id	integer (FK → users)	The mahasiswa who found it
deskripsi	text	How/where the item was found
lokasi_sekarang	IPBLocation	Where the item is currently stored
foto_bukti	string[]	Evidence photo URLs. Default: []
status	FoundReportStatus	PENDING APPROVED REJECTED. Default: PENDING
catatan_admin	text (nullable)	Admin note
kode_pengambilan	string (nullable)	Pickup code for item owner, set on approval
created_at	datetime	Submission time

10.5 ItemMatch

Field	Type	Description
id	integer (PK)	Primary key
found_item_id	integer (FK → items)	The FOUND item in the match pair
lost_item_id	integer (FK → items)	The LOST item in the match pair
alasan_match	text	System explanation: why this match was detected
status	MatchStatus	PENDING CONFIRMED REJECTED. Default: PENDING
catatan_admin	text (nullable)	Admin note on confirm/reject
created_at	datetime	When the match was auto-detected

10.6 Notification

Field	Type	Description
id	integer (PK)	Primary key
user_id	integer (FK → users)	Recipient user ID
judul	string	Short notification title
pesan	text	Full notification message
is_read	boolean	Read status. Default: false
created_at	datetime	Creation timestamp

10.7 PointLog

Field	Type	Description
id	integer (PK)	Primary key
user_id	integer (FK → users)	User who earned/lost points
jumlah	integer	Point delta: positive=earned, negative=deducted
alasan	string	Human-readable reason for the transaction
created_at	datetime	Transaction timestamp

11. Business Logic & Rules

11.1 Points System

Points incentivize students to report and return found items. Points are awarded automatically when admin approves claims/reports or confirms matches.

Event	Who Earns	Points
Claim APPROVED	Finder (reporter of the FOUND item)	+10 poin
Found Report APPROVED	Reporter (mahasiswa who found and submitted the report)	+10 poin
Match CONFIRMED	Finder (reporter of the FOUND item in the match)	+10 poin

11.2 Auto-Match Engine

Triggered automatically when a new item is created (POST /items).

Condition	Score Added
Same kategori (category)	+3 points
Overlapping ItemLocation (lokasi_ditemukan_list / lokasi_kemungkinan_list)	+2 points
Match created if total score >= 3	→ ItemMatch with status=PENDING

alasan_match example: "Kategori sama: dompet | Lokasi berdekatan: Fakultas Ekonomi dan Manajemen (FEM)"

11.3 Item Expiry

A background scheduler runs every 24 hours (APScheduler) and marks items as EXPIRED if they have not been resolved and the current time has passed expired_at.

• expired_at is automatically set to created_at + 30 days when the item is created. • Items with status CLOSED are never expired. • Expired items remain in the database but are filtered out of active searches by default. • expired_at is visible in both ItemOut and ItemOutPublik for frontend countdown display.

11.4 Pickup Codes (kode_pengambilan)

A pickup code is a 6-character alphanumeric token generated when an admin approves a Claim or Found Report. It serves as a physical handover verification mechanism.

• Generated using: secrets.token_hex(3).upper() → e.g. 'A3F9C2' • The code is given to the item owner (claimant) via notification. • The finder/admin presents this code when physically handing over the item. • Admin can retrieve the code anytime via GET /claims/{id}/kode or GET /found-reports/{id}/kode. • The code is stored in the claim/found_report record and never changes after generation.

Quick Reference — All Endpoints

Method	Endpoint	Auth	Description
POST	/auth/login	Public	Login → get JWT

Method	Endpoint	Auth	Description
GET	/auth/me	Any user	Get own profile
GET	/items	Public	List items with filters
GET	/items/admin/all	Admin	All items (full detail)
GET	/items/{item_id}	Public	Item detail
POST	/items	Mahasiswa	Report lost/found item
POST	/items/{item_id}/foto	Mahasiswa	Upload item photo
POST	/items/{item_id}/bukti-kepemilikan	Mahasiswa	Upload ownership proof
DELETE	/items/{item_id}	Admin	Delete item
POST	/claims	Mahasiswa	Submit claim
POST	/claims/{claim_id}/bukti	Mahasiswa	Upload claim evidence
GET	/claims/me	Mahasiswa	My claims
GET	/claims	Admin	All claims
PATCH	/claims/{claim_id}/approve	Admin	Approve claim
PATCH	/claims/{claim_id}/reject	Admin	Reject claim
GET	/claims/{claim_id}/kode	Admin	Get pickup code
POST	/found-reports	Mahasiswa	Submit found report (multipart)
POST	/found-reports/{report_id}/foto	Mahasiswa	Upload evidence photo
GET	/found-reports/me	Mahasiswa	My found reports
GET	/found-reports	Admin	All pending found reports
GET	/found-reports/{report_id}/kode	Admin	Get pickup code
PATCH	/found-reports/{report_id}/approve	Admin	Approve found report
PATCH	/found-reports/{report_id}/reject	Admin	Reject found report
GET	/matches	Admin	List pending matches
PATCH	/matches/{match_id}/confirm	Admin	Confirm match
PATCH	/matches/{match_id}/reject	Admin	Reject match
GET	/notifications	Any user	Get notifications
PATCH	/notifications/read	Any user	Mark all as read
GET	/leaderboard	Public	Top 10 leaderboard
GET	/leaderboard/me/riwayat	Mahasiswa	My points history

End of API Contract — IPB Lost & Found System v1.0.0

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